13.5.16(m)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

April - May 2016/ ~~Nov - Dec 2016~~

Max. Marks: 100 | **Duration: 3 hrs**
Class: SE | Semester: IV
Name of the Course: Theoretical Computer Science | Branch: CMPN
Course Code: **UCEC405**

Instructions:

(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1	**Answer any five questions**	**25**
Q 1 (a)	Design a FSM over $\sum\{0,1\}$ in which input is valid if it contains atmost three 1's	
Q 1 (b)	Differentiate between Moore and Mealy Machines	
Q 1 (c)	List the Kleen's Closure Properties	
Q 1 (d)	Design a DFA to accept strings ending with "aa" or "bb" over $\sum\{a,b\}$	
Q 1 (e)	What are the variants of Turing Machine? Explain in brief.	
Q 1 (f)	Using pumping Lemma show that the language $L = \{1^n01^n / n\geq1\}$ is not regular	
Q2 (a)	Construct a PDA accepting the language $L = \{a^nb^n \mid n>0\}$	**10**
	OR	
	Design a PDA corresponding to the grammar $S \rightarrow aSa \mid bSb \mid \varepsilon$	**10**
Q2 (b)	Construct a Turing Machine to check for well-formed parentheses	**10**
Q 2 (c)	Discuss Parsing and Ambiguity with an example	**05**
Q 3 (a)	Design a Turing Machine which accepts any language L over {a,b} which has equal numbers of a's and b's	**10**
	OR	
	Construct a TM to accept the language $L = \{wcw^R \mid w \in \{0, 1\}$ and w^R means reverse of w}. Trace the sequence of moves corresponding to input string 110c011.	**10**
Q 3 (b)	Convert the following regular expression to DFA :- (01 + 11)*	**10**
Q 3 (c)	Discuss briefly applications of Regular Expressions	**05**
Q 4 (a)	Explain CNF and GNF. Convert the following grammar into GNF. S->ASB|a|bb A->aSA|a B->SBS|bb	**10**
	OR	
	What is MyHill Nerode Theorem? Where can it be used? Discuss any one application of MyHill Nerode Theorem	**10**

Q 4 (b)	Explain relation between FSM, PSM and TM	10
Q 4 (c)	Design a FSM that checks if the given number is odd	05

OR

Q 4(b).	Design Moore and Mealy Machine to find 2's complement of a binary number	10